

Danilo Augusto Silva

POST-DOCTORAL RESEARCHER · PHD IN PHYSICAL OCEANOGRAPHY

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Education

University of São Paulo	<i>São Paulo, SP</i>
PHD IN PHYSICAL OCEANOGRAPHY	2019 - 2024
University of São Paulo	<i>São Paulo, SP</i>
MSC IN PHYSICAL OCEANOGRAPHY	2017 - 2019
University of São Paulo	<i>São Paulo, SP</i>
BSC IN OCEANOGRAPHY	2010 - 2016

Professional Experience

2024-2025	Post-doctoral Researcher , Plankton System Lab, University of São Paulo, SP
2023-2023	Research Student Visitor , University of Massachusetts - School for Marine Sciences and Technology (SMAST)
2019-2024	Environmental Data Analyst , Oracle Meteorology, São Paulo, SP
2017-2024	Graduate Research Assistant , Coastal Hydrodynamics Lab, University of São Paulo, SP
2015-2016	Undergraduate Research Assistant , Coastal Hydrodynamics Lab, University of São Paulo, SP
2014-2015	Undergraduate Research Assistant , Oceanographic Data Lab, University of São Paulo, SP
2011-2012	Undergraduate Research Assistant , Satellite Oceanography Lab, University of São Paulo, SP
2010-2011	IT Auxiliar , Media Coordination of University of São Paulo, SP

Research Experience

Postdoctoral Research, University of University of São Paulo	2024 - 2025
- Investigating physical-biological interactions around seamounts and the Amazon River plume and their roles on the particulate organic carbon (POC) vertical fluxes; - Using observational dataset (UVP, VMP and CTD) and numerical modeling (ROMS, Parcels); - Performing hydrographic, meteorological and ecological data analysis on other projects; - Funded by University of São Paulo Foundation (FUSP). Two manuscripts currently in preparation.	
Research visitor, University Utrecht - Institute for Marine and Atmospheric research Utrecht (IMAU)	2025 - 2025
- Explored how Lagrangian analysis and modeling can improve our understanding on carbon sequestration pathways associated with Equatorial river plumes; - Developing vertical sinking module for Parcels (lagrangian model), including remineralization rates to understand the particulate organic carbon (POC) fate associated with the Amazon River Plume; - Visit funded by University of São Paulo Foundation, through AtlantECO project.	
Research student visitor, University of Massachusetts - School for Marine Sciences and Technology (SMAST)	2023 - 2023
- Investigated pathways of salty and warm subsurface intrusions (Salinity Maximum) on the Southwestern South Atlantic; - Using ARGO and Slocum Glider dataset as well as lagrangian modeling to investigate the interaction of mesoscale features and wind-driven circulation; - Award funding from University of São Paulo Foundation. Manuscript in revision.	
Graduate Student (PhD), University of São Paulo	2019 - 2024
- Investigated subsurface salty and warm intrusions at the southeastern Brazilian continental shelf, using a historical dataset and detection algorithms; - Described the main patterns of these features along the shelf, relating their properties to river runoff and shelf's with. - Funded by University of São Paulo Foundation (FUSP). One peer-reviewed published manuscript.	

Graduate Student (MSc), University of São Paulo

2017 - 2019

- Investigated persistent events of atmospheric blocking influence over the continental shelf hydrodynamics;
- Activated heat fluxes module on numerical model (ECOM), to better represent the sea surface temperature variability under these conditions;
- *Funded by CNPq. One peer-reviewed published manuscript.*

Onboard Research Assistant, Impacts of the Southwestern Atlantic on the South American Climate - The Oceanic Importance

2016

- Activities: Deployment array of moored current meter line crossing the Brazil Current.

Onboard Research Assistant, Challenger Project - South Atlantic Circumnavigation

2015

- Activities: Testing glider's navigation system and releasing the RU29 - Rutgers glider. More information at <https://challenger.marine.rutgers.edu>

Publications

PUBLISHED

- Santos, A. D., **Silva, D. A.**, Sasaki, D. K., Dottori, M. (2025). Microplastic Hotspots on a Tropical Estuarine-Bay System. Em SciELO Preprints. <https://doi.org/10.1590/SciELOPreprints.12903>
- Dottori, M., Sasaki, D.K., **Silva, D. A.**, Birocchi, P., do Nascimento, R.F. (2025). Physical Oceanography of the Brazilian Continental Shelf. In: Sumida, P.Y.G., Mies, M. (eds) Oceanography, Biodiversity, Fisheries and Conservation of Brazilian Continental Shelf Habitats. Brazilian Marine Biodiversity. Springer
- Sasaki, D. K.; **Silva, D. A.**; Del-Giovanino, S. R.; Dottori, M.; Silveira, I. C. A.; Belo, W. C.; Martins, R. P. (2024). Hydrographic climatology of the South Brazil Bight continental shelf and slope. Theoretical and Applied Climatology.
- Moreira, D. L., ..., **Silva, D. A.**, and Souza, S.H. (2023). Multidisciplinary scientific cruises for environmental characterization in the Santos Basin—methods and sampling design. Ocean and Coastal Research.
- Silva, D. A.**, Sasaki, D. K., Dottori, M., da Silveira, I. C. A., Belo, W. C., and Martins, R. P. (2023). An inventory of salty and warm subsurface intrusions in the South Brazil Bight. Continental Shelf Research.
- Birocchi, P., **Silva, D. A.**, Sasaki, D. K., and Dottori, M. (2023). Estimating river discharge from rainfall satellite data through simple statistical models. Theoretical and Applied Climatology.
- Dottori, M., Sasaki, D. K., **Silva, D. A.**, Del-Giovanino, S. R., Pinto, A. P., Gnamah, M., ... and Moreira, D. L. (2023). Hydrographic structure of the continental shelf in Santos Basin and its causes: The SANAGU and SANSED campaigns (2019). Ocean and Coastal Research.
- Mendes, C. B., Cortez, T., Santos, C. S., Sobral-Souza, T., Santos, A. D., Sasaki, D. K., **Silva, D. A.**, Dottori, M. and Andrade, S. C. (2022). Seascape genetics in a polychaete worm: Disentangling the roles of a biogeographic barrier and environmental factors. Journal of Biogeography.
- Silva, D. A.** and Dottori, M. (2021). The atmospheric blocking influence over the South Brazil Bight during the 2013–2014 summer. Regional Studies in Marine Science.

IN REVIEW

- P. Birocchi, Souza W. O.; Braz, Caio; **Silva, D. A.**; Frazão, L. R.; Barrera-Alba, J.; Lessa, G. C. High-resolution temporal dynamics of diatoms in a large and well-mixed tropical estuary. Submitted to Marine Environmental Research.
- Vilanova, G; Zani-Teixeira, M. L.; Katsuragawa, M.; Santos, U. C.; Dottori, M.; **Silva, D. A.**; Silveira, I. C. A.; Amorim, J. P. M.; Brandini, F. P.; Namiki, C. Water mass influence on the distribution of larval mesopelagic fish (Sternoptychidae) in the South Brazil Bight (23oS - 28oS). Submitted to Journal of Marine Systems.
- Silva, D. A.**; Gangopadhyay, A.; Sasaki, D. K. and Dottori, M. Silveira, I. C. A., Belo, W. C., Martins, R. P. Evidence of Synoptic Cross-shelf Pathways of Mid-depth Salinity Maximum Intrusions in the South Brazil Bight. Submitted to Journal of Geophysical Research: Oceans

IN PREPARATION (SUBMISSION WITHIN 3 MONTHS)

- Silva, D. A.**; Sasaki, D. K. and Dottori, M. Physical drivers of mid-depth Salinity Maximum intrusions in the South Brazil Bight (SBB).

Silva, D. A.; Santos, G.; Penninck, S.; Guidi, L.; Elineau, A. Calil, P.; Lopes, R. Flow-topography interactions drive zooplankton abundance and carbon export to depth along the Vitória-Trindade Seamount Chain (Southwestern South Atlantic)

Penninck, S.; Lopes, R.; **Silva, D. A.;** Santos, G.; Tuchen, F. P.; Baudena, A.; Stemmann, L.; Krahmann, G.; Begler, C.; Brandt, P.; Kiko, R. Influence of upper-ocean physical processes on the carbon flux variation in the mesopelagic equatorial Atlantic

Presentations

ORAL PRESENTATIONS

Oral presentation at **Parcels 10-year Anniversary**, (Utrecht, Netherlands, 2025).

Oral presentation at **AtlantECO Final Scientific Conference**, AtlantECO (Ponta Delgada, Azores, 2025).

Oral presentation at **AtlantECO Final Scientific Conference**, AtlantECO (Ponta Delgada, Azores, 2025).

Oral presentation at **Latin-American Physics of Estuarine and Coastal Ocean**, LAPECO (Panama City, Panama, 2023).

Virtual poster presentation at **Ocean Sciences Meeting** (2022).

Virtual presentation at **I Early Career Symposium on Biogeochemical Modeling** (2021).

Oral presentation at **Latin-American Physics of Estuarine and Coastal Ocean**, LAPECO (Florianópolis, Santa Catarina, 2019).

POSTER PRESENTATIONS

Poster presentation at **Gordon Research Conference - Coastal Dynamics** (Smithfield, Rhode Island, 2023).

Poster presentation at **Western Boundaries Current and Shelf interactions** (Savannah, Georgia, 2023).

Poster presentation at **Symposium on Waves, Tides, Oceanic Engineering and Satellite Oceanography**, OMARSAT (Arraial do Cabo, Rio de Janeiro, 2019)

Poster presentation at **Symposium on Waves, Tides, Oceanic Engineering and Satellite Oceanography**, OMARSAT (Arraial do Cabo, Rio de Janeiro, 2017)

Poster presentation at **10th International Workshop on Modeling the Ocean**, IWMO (Santos, São Paulo, 2017).

Teaching Experience

Invited Lecturer, Volunteer, OceanHackWeek 2023

- *Course:* Introduction to multidimensional data analysis with xarray.
- *Responsibilities:* Preparing a 2-hour single class for participants on Oceanhackweek.

Helper, Volunteer, OceanHackWeek 2022

- *Responsibilities:* Helping participants with Git and Python during the event.

Lecturer, Volunteer, University of University of São Paulo 2020, 2021, 2025

- *Course:* Python for Oceanographers
- *Responsibilities:* Introducing undergraduate students to Python, preparing and presenting classes independently, taking questions and group sessions for guidance on best practices and data analysis methods for students own projects.

Teaching Assistant, University of University of São Paulo 2019

- *Course:* Introduction to Physical Oceanography
- *Responsibilities:* Preparing assignments and having group session for taking questions.

Teaching Assistant, University of University of São Paulo 2019

- *Course:* Coastal and Estuarine Physical Oceanography
- *Responsibilities:* Preparing assignments and having group session for taking questions.

Skills and Activities

SKILLS

- Data Science **Proficient in Python**, using scientific stack, Bash Scripting, Linux, SQL, Version Control (Github), Data processing pipelines using workflow orchestration (Apache Airflow), Containerization/Encapsuling (Docker), Serverless services (AWS lambda) and API development (Django and Rest).
- Research **Numerical modeling**, using ROMS, Delft3D, ECOM/POM, HEC-RAS. Lagrangian analysis (Parcels, Opendrift), Statistical Analysis (parametric and non-parametric), **Observational Data Analysis** (CTD, ARGO floats, SlocumGlider, UVP), Data Visualization (Python), and Scientific Writing.

ACTIVITIES

- Peer Review **Journals that I reviewed for**, Continental Shelf Research, Journal of Marine Systems
- Event **Organization of National Week of Oceanography**, academic and sportive event for more than 400 undergraduate students, with poster presentation sessions, courses and talks during the week.
- Event **Organization of donation campaigns**, for bone marrow and underprivileged communities.
- Leadership **Student Society**, responsible for the finances in 2011 and sport modality manager (rugby, 2012 to 2014).
- Sport **National and State League 1 Champion (Poli Rugby, 2018 and 2019)**, state League 2 Champion (Urutu Rugby, 2023), 2024's Urutu Rugby Captain